

EDUCATION

Year	Degree/Certificate	Institute	CGPA/%
2023–Present	B.Tech — AI & Data Science	Madras Institute of Technology, Anna University	8.21 / 10.0
2023	Higher Secondary	Veveaham Matric Hr. Sec. School, Dharapuram	91.17 %

EXPERIENCE

- Ticketing Management System (ITSM Platform)

Database Design & QA

2026

Student Consultancy Project

– Contributed to designing a monolithic application following Domain-Driven Design (DDD) principles.

– Designed the PostgreSQL schema for a multi-module ITSM platform, covering tickets, agents, roles, queues, and SLA rules.

– Defined SLA logic — region-based working hours, escalation timers, and breach tracking — backed by normalized DB tables.

– Authored a Knowledge Base (KB) structure supporting agent self-service and faster ticket resolution.

– Tested REST APIs via Postman against Swagger specs; performed functional and regression testing across routing, permissions, and notifications.

PROJECTS

- TEKHORA’26 Symposium Website

React, Firebase, Node.js

2026

– Developed the official website for the TEKHORA’26 college symposium, enabling registrations and secure online payments.

– Built backend services with Firebase Cloud Functions and Node.js; integrated Cashfree payment gateway for transactions.

– Deployed React frontend via Firebase Hosting; managed registrations and event data with Firestore; added OTP verification via Resend.
- Attention-Enhanced LSTM Autoencoder for ADS-B Monitoring

Deep Learning Techniques, Mini Project

2026

– Implemented and extended the LSTM-Transformer Fusion VAE (LTF-VAE) framework from IEEE Doc. 11181846 for unsupervised anomaly detection in ADS-B trajectory data.

– Trained baseline LSTM-VAE and Transformer LSTM-VAE models on the OpenSky ADS-B dataset via a sliding-window pipeline (length 10) over seven trajectory features.

– Achieved 95.40% accuracy and 0.898 AUC-ROC with the LSTM-VAE baseline using reconstruction-error anomaly scoring.

– Built an interactive Flask dashboard (Chart.js/D3.js) for model comparison, trajectory visualization, and threshold-adjustable exploration.
- Intelligent On-Board Weighing (OBW) Kit – AI-Driven Vehicle Mass Estimation

Student Innovative Project (SIP), Python, TensorFlow/Keras, Raspberry Pi 4

2026

– Trained a GRU-based regression model (TensorFlow/Keras) on filtered ECU data to estimate vehicle mass in real time from OBD-II parameters, per AIS-205 guidelines.

– Ran experiments across full (17-feature) and OBD-II-only (9-feature) configurations, achieving 1.57 kg MAE.

– Converted the model to int8/float16-quantized TensorFlow Lite and deployed it on a Raspberry Pi 4 with sub-100 ms latency.

– Built the end-to-end on-device pipeline: OBD-II acquisition, GRU inference, LCD output, and SD-card logging.

TECHNICAL SKILLS

- Programming Languages: Python, Java, C, C#, JavaScript
- Web Development: HTML, CSS, React, Node.js, Express.js, ASP.NET
- Databases: MongoDB, MySQL, PostgreSQL, Firebase Firestore, Redis
- Backend/Messaging: RabbitMQ, REST APIs, Postman, Swagger
- Machine Learning: Machine Learning, Deep Learning, Computer Vision
- Tools: Git, Docker, Firebase, Kaggle, NodeMailer, Cashfree Payment Gateway
- Data Analysis: NumPy, Pandas, Matplotlib

LANGUAGES

- English: Conversational
- Tamil: Native

CERTIFICATIONS

- Programming in Java — NPTEL
- Design Thinking — NPTEL
- Artificial Intelligence Fundamentals — IBM SkillsBuild